$$4(x^2 - 4x + 8) = (x^2 - 16x + 80)$$

$$4x^2 - 16x + 32 = x^2 - 16x + 80$$

$$3x^2 - 48 = 0$$

$$3x^2 = 48$$

$$x^2 = 16$$

$$x = \pm 4$$

As the point P lies on x-axis (positive side), its x- coordinate cannot be -4.

Hence the coordinates of P is$(4, 0)$

Example 5.11

Show that (4, 3) is the centre of the circle passing through the points (9, 3), (7,–1), (–1,3). Also find its radius.

Solution

Let $P(4, 3)$, $A(9, 3)$, $B(7, -1)$ and $C(-1, 3)$

If P is the centre of the circle which passes through the points A, B, and C, then P is equidistant from A, B and C (i.e.) $PA = PB = PC$

By distance formula,

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$AP = PA = \sqrt{(4-9)^2 + (3-3)^2} = \sqrt{(-5)^2 + 0} = \sqrt{25} = 5$$

$$BP = PB = \sqrt{(4-7)^2 + (3+1)^2} = \sqrt{(3)^2 + (4)^2} = \sqrt{9+16} = \sqrt{25} = 5$$

$$CP = PC = \sqrt{(4+1)^2 + (3-3)^2} = \sqrt{(5)^2 + 0} = \sqrt{25} = 5$$

$PA = PB = PC = 5$, Radius $= 5$

Therefore P is the centre of the circle, passing through A, B and C

Exercise 5.2

1. Find the distance between the following pairs of points.
   (i) (1, 2) and (4, 3) (ii) (3,4) and (– 7, 2)
   (ii) (a, b) and (c, b) (iv) (3,– 9) and (–2, 3)

2. Determine whether the given set of points in each case are collinear or not.
   (i) (7,–2),(5,1),(3,4) (ii) $(a,-2)$, $(a,3)$, $(a,0)$

3. Show that the following points taken in order form an isosceles triangle.
   (i) A (5,4), B(2,0), C (–2,3) (ii) A (6,–4), B (–2, –4), C (2,10)

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